

QuantWiFi Agent Requirements For Turk Telekom Gateways

1. QuantWiFi Agent

QuantWiFi Agent(wifiSpeed) is an application downloaded into “Gateway/Access Point” at start-up and after every reboot. Agent is responsible for collecting Wi-Fi metrics, initiate Wi-Fi measurements and posting results to Turk Telekom Data Center for further analysis.

Latest QuantWiFi Agent (“wifiSpeed”) will be served from Turk Telekom file server. File Server typical URL will look like this “<https://xx.turktelekom.com.tr>”. The location of the agent on the server will be “[fileserver/QuantWiFi/VENDOR_MODEL/wifiSpeed](#)”. **VENDOR** is the code name used by Turk Telekom to refer to fileserver directory for that specific vendor, and the **MODEL** is the code name referring to modem product of concern from the vendor. Both **VENDOR**, **MODEL** code names should be validated with Turk Telekom before use.

QuantWiFi agent URL location is subject to modification via TR-069 (see below). Turk Telekom Gateways are supposed to download the agent from Turk Telekom fileserver at the location depending on the VENDOR and MODEL name.

The QuantWiFi agent should be saved into /var/tmp directory and run from there using ‘wifiSpeed’. If for any reason QuantWiFi agent fails, host device should download from remote Turk Telekom server, save it into /var/tmp directory and run again.

2. QuantWiFi Loader

Following ‘QuantWiFi Loader’ high level script or similar code piece can be used in the background. The ‘QuantWiFi Loader’ script will behave according to Tr069 parameter values. Refer to TR-069 Parameters section for details.

```
#!/bin/sh
while true; do \
    sleep 10 && \
    wifipid=$(pidof wifiSpeed) && \
    kill -9 $wifipid && \
    rm -f /tmp/wifiSpeed && \
    wget URL -O /tmp/wifiSpeed && \ (or curl -cacert X URL -output /tmp/wifiSpeed)
    chmod +x /tmp/wifiSpeed && \
    param=KEY /tmp/wifiSpeed; \
done
```

Actual Loader is more complex than what is presented above. QuantWifi can share its Loader Script prepared for different platforms/vendor devices if requested. QuantWiFi Agent will be served together with its signed version and will be verified in the device before attempting to run the agent.

Host device is responsible keeping the 'QuantWifi Loader' script up all the time. In any circumstance, if the 'QuantWifi Loader' script stops running, host device should restart the script.

3. TR-069 Parameters

'QuantWifi Loader' script or similar implementation behavior will be remotely managed by the following TR-069 parameter set reserved by Turk Telekom and needs to be implemented in the Gateway:

1. **InternetGatewayDevice.X_TTNET.Configuration.QuantWiFiAgent.URL (will be referred as 'URL')**
 1. Parameter Type: string
 2. Parameter Attribute: W (writeable)
 3. Parameter Default Value: An example is provided below and must be verified with QuantWifi and TurkTelekom.
https://acs-fileserver.turktelekom.com.tr/filesserver/QuantWiFi/VENDOR_MODEM/wifiSpeed
 4. This parameter set the URL from which QuantWiFi agent will be downloaded by the 'QuantWiFi Loader' script
2. **InternetGatewayDevice.X_TTNET.Configuration.QuantWiFiAgent.KEY (will be referred as 'KEY')**
 1. Parameter Type: string
 2. Parameter Attribute: Write Only (not readable; fail in case read attempt)
 3. Parameter Default Value: NONE
 4. This parameter will be input as a parameter to QuantWifi agent.
3. **InternetGatewayDevice.X_TTNET.Configuration.QuantWiFiAgent.State (will be referred as 'STATE')**
 1. Parameter Type: string
 2. Parameter Attribute: R (Read Only)
 3. Parameter Default Value: 'NONE'
 4. This parameter has two values 'RUNNING' and 'NONE'. 'STATE' parameter is used to indicate the 'QuantWiFi Loader' script status. 'RUNNING' indicates 'QuantWiFi Loader' script is added into boot sequence of the host device (i.e. added into rc scripts execution sequence) and at the same time 'QuantWiFi Loader' script started running in the background. 'NONE' indicates the 'QuantWiFi Loader' script stopped execution and removed from the boot sequence (i.e. removed from rc scripts execution sequence).
4. **InternetGatewayDevice.X_TTNET.Configuration.QuantWiFiAgent.Enable**
 1. Parameter Type: Boolean
 2. Parameter Attribute: W (writeable)
 3. Parameter Default Value: FALSE
 4. When set to TRUE, 'QuantWiFi Loader' script is added into boot sequence of the Gateway (i.e. added into rc scripts so that it will start executing every time Gateway started/reboot), and runs itself. The loader script will download QuantWiFi agent from 'URL' into device tmp directory and run it from there. 'STATE' parameter will be switched to 'RUNNING'.
 5. When set to FALSE, 'QuantWiFi Loader' script stops running and removed from boot sequence. 'STATE' switched to 'NONE'. Once stopped, loader script as well as QuantWiFi agent ('wifiSpeed') and its children should be unloaded from memory.

Tr-069 parameters listed above must functioning atomically and should satisfy the following:

1. If "STATE" of the agent is "NONE", setting 'Enable' parameter to TRUE via Tr-069 will start agent as described above, with "KEY", "URL" parameter values already set.
2. If "STATE" of the agent is "NONE", setting 'Enable' parameter to TRUE while updating "KEY" and "URL" parameters via Tr-069 atomically (in one shot), will start agent as described above but with updated/new "KEY", "URL" parameter values.
3. If "STATE" of the agent is "RUNNING" then any change in "KEY" or "URL" parameters via Tr-069 should trigger QuantWiFi Agent restart like "Enable" parameter to set to TRUE, but with updated/new values

4. Key Elements

QuantWiFi agent relies on following key elements on the host device to run properly:

1. WGET or CURL application in the CPE device with certificate validation support for downloading from TLS/SSL protected file server.
2. A set of certificates placed under /etc/qw/certs/ directory.
3. Openssl application with "dgst" option support available on the CPE device or alternatively Quantwifi provided 'qwutil' application implementing "openssl dgst" function for agent signature check inside CPE.
4. Access to IP address and friendly device names in DHCP protocol of the associated clients. Typically access to DHCP Lease text file is enough. If this is not available, an API should be available to query DHCP NAME and IP ADDRESS of all associated clients using its mac address. The API output should be clearly documented and should cover all the potential corner cases. QuantWiFi do not prefer text output since parsing text output is error-prone and require iterations usually.
5. DHCP and WiFi SSID names that support multi-languages such as English and Turkish character sets. For that matter Unicode encoding (specifically UTF-8) support is must. If any other character coding is used (BIG5, Extended ASCII character pages), it must be shared with QuantWiFi aspart of the documentation
6. If Auto Channel selection mechanism supported by the platform, details should be provided to QuantWiFi
7. The kernel should have "epoll" support build in as well as /sys/ directory structure.
8. In case of hw accelerators used on certain interfaces, the accelerator network statistics should be combined with interface statistics and be visible using "ifconfig" or printing /proc/net/dev.
9. Access to SN (Serial No) and Device FW Version (fw version) via command line
10. Robust access to wireless driver:
For Broadcom based platforms this refer to "wl"
For MTK based devices this refers to following:
 1. List of IOCTLs :
IOCTL: SICOGIWNAME
IOCTL: RTPRIV_IOCTL_GET_MAC_TABLE
IOCTL: RTPRIV_IOCTL_GSITESURVEY
IOCTL: RTPRIV_IOCTL_GET --- Parameter: stainfo
IOCTL: RTPRIV_IOCTL_GET--- Parameter: devinfo
IOCTL: RTPRIV_IOCTL_GET --- Parameter: SSID
 2. MTK/Econet patch referred to as "patch_for_supporting_190049_QuantWiFi_TT"
 3. Future patch from MTK/Econet
11. CPE device Toolchain: to generate QuantWifi Agent binary for the CPE device
12. Root access to device to test and validate QuantWifi Agent functionality on the CPE device.